14.3.1 The cache tag helper

The cache tag helper enables you to cache regions of a Razor page in memory on the server. Typical uses for this helper are for *view components* or partial views that require relatively expensive database or web service calls, but where the content doesn't change very often. Unlike most other tag helpers, the cache tag helper doesn't target a standard HTML element. It targets the `<cache>` tag, which is not rendered to the browser and doesn't appear in the HTML source either.

The cache tag helper uses an in-memory-based store for content. This is volatile and can be cleared unexpectedly for any number of reasons, including app pool recycling, server restarts, low-memory pressure, and so on. The cache tag helper is not intended to be used for reliable long-term storage.

To demonstrate the value of the cache tag helper properly, you will create a view component that will produce the data-driven content currently being produced by the home page. If you recall from chapter 3, view components are similar to partials in that they are designed to encapsulate regions of pages, but they differ in that they are designed to work with services and generate their own data. View components are classes that derive from `ViewComponent` and implement an `Invoke` method that returns an `IViewComponentResult` or an asynchronous version that returns a `Task<IViewComponentResult>`. They must also have `ViewComponent` as a suffix to their name or be decorated with the `[ViewComponent]` attribute. Listing 14.19 shows a class named `CitiesViewComponent` (placed in a folder named ViewComponents) that follows these rules. It takes the `ICityService` as a dependency via its constructor and uses it to create a model consisting